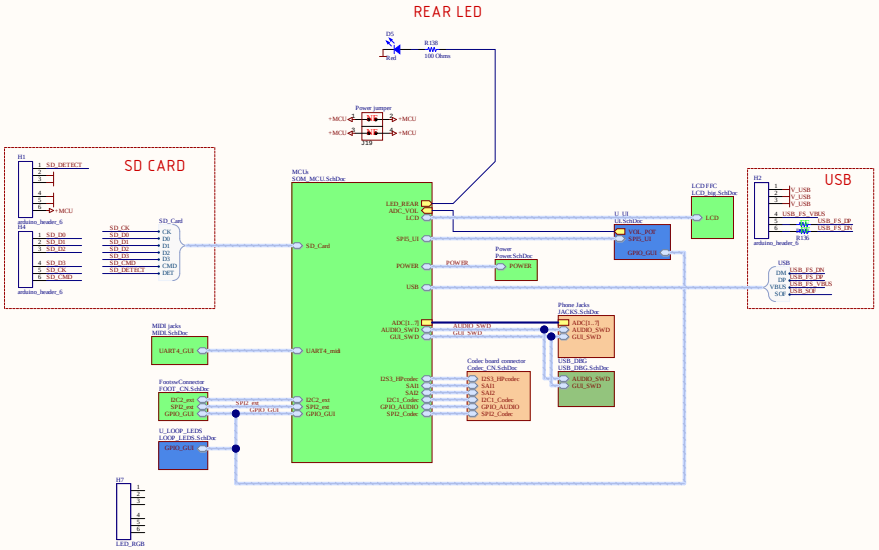
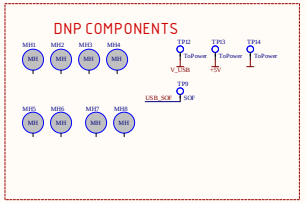






Created masks for FLASH_ nets cmosis, masked unused headers
- resolveScan:
InNet(FLASH_D3,FLASH_D2,FLASH_D1,FLASH_D0,FLASH_CLK)

Даже говорил, как кнопки расходные. футпанчи получается на гпио пойдут

Добавил шлейфы стоячие на макету для модуля и футсвичей, и надо передвинуть джеки на плате одна

Сделать платы XLR, объединить плату джеков и сместе вместе ребольты их шлейфов, чтобы были параллельно рять на плате модуля

Мидю и джеки на подключены

=== FIXING LIST ===
> reverse SD card
> LED regs reset - wrong
> input regs NRST - do i need it?
> shorter rear buttons model
> More resistance on LEDs
> Redesign SWD debug sockets
> Redesign mezzanine situation
> Think about slide-in FFC instead of cables
> remeasure LCD exact position

Recalculate R- resistors for DC-DC for 5V output
Change barrel power cups to tantal (for low profile)
Add OVP for +5V to protect registers from ripple
Add D00H for soft-start and current-limiting of USB

Change LED resistor networks to 510 ohm
Encoder goodliier RC filters - update sch to 5100R/0.1uF
Add ESD protection on SD card lines
Add SD detect hardware pullup
Add LCD lines current limiting resistors 22R

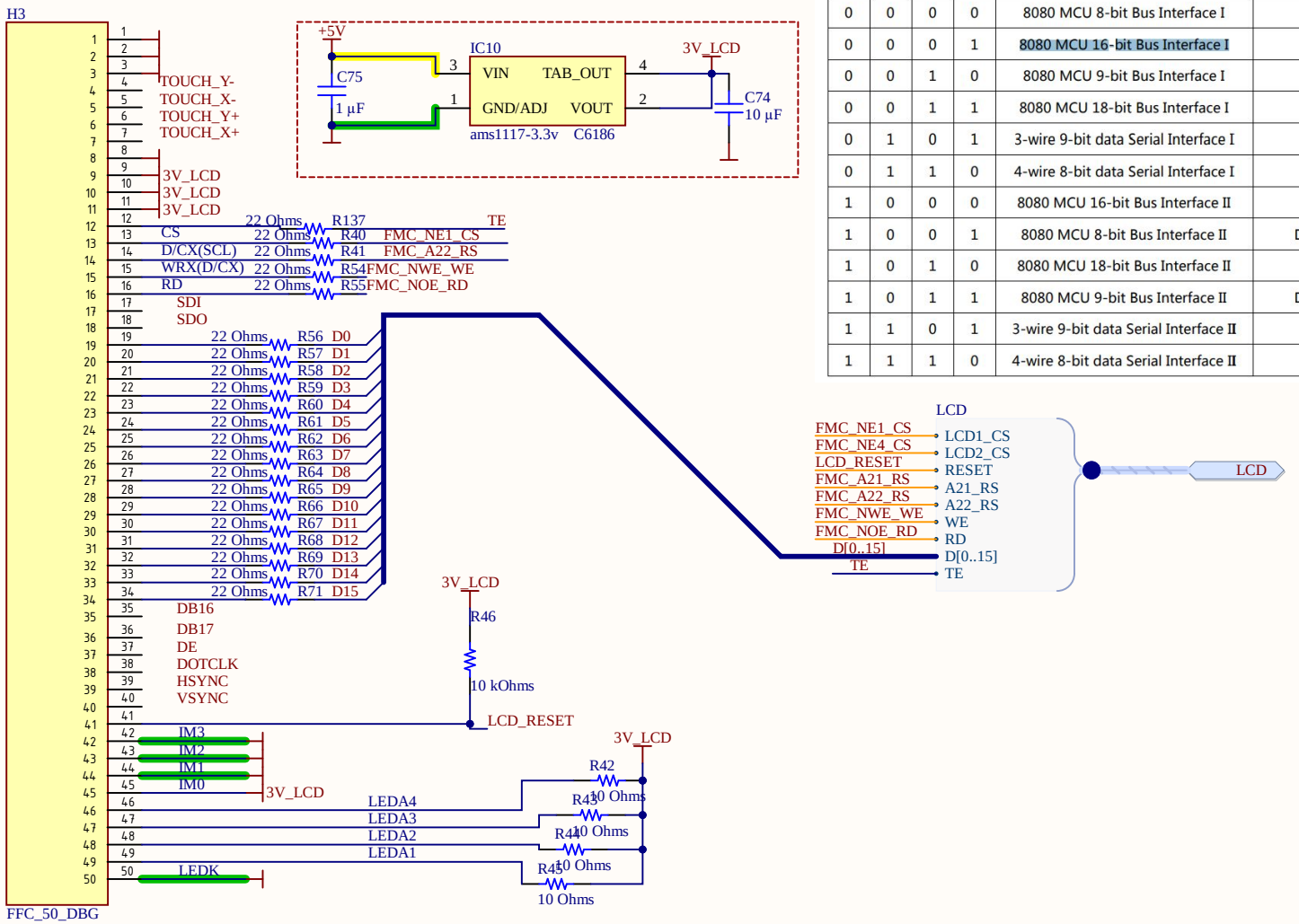
Tutorial:
> for connectors, add System -> Connector Parameter to connectors of multibuses
> make int lib in all mh projects and in mh sch in every connection of type "cable" select connector component from that int libs, then go to connection properties, click no connection and then back for connections to reppear and update
> in mh sch - design -> connection manager - useful - show changes only option - revert conflict or swap pins, then update child projects for back ecn

> Select menu on S key in PCB editor
> I have disabled Active Route - need another license or altium version?

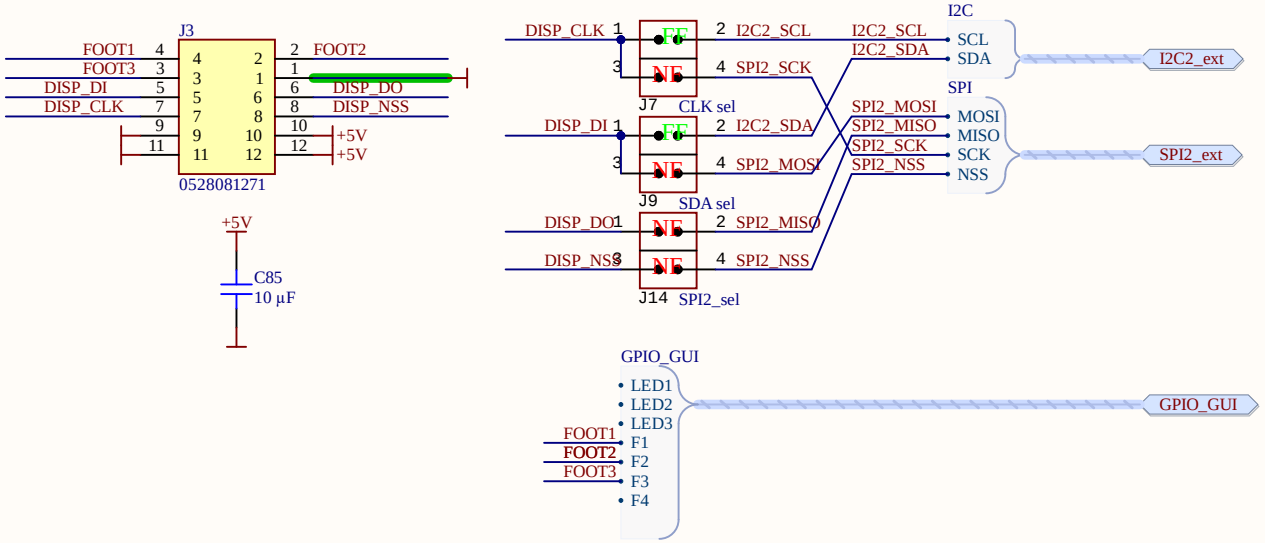
Title			
Size	Number		Revision
A1			
Date:	10.29.2026	Sheet of	
File:	C:\Users\...\hd_dsn_approves.SchDoc	Drawn By:	

Note 1: Select the MCU interface Mode

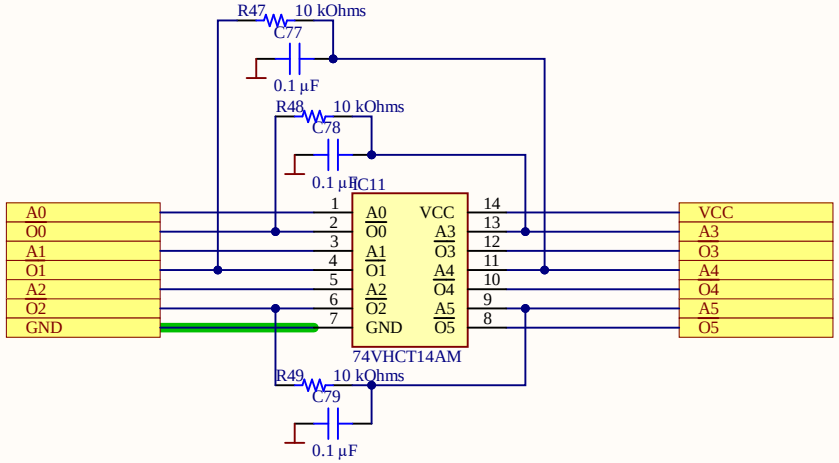
IM3	IM2	IM1	IM0	MCU-Interface Mode	Pin in Use	
					Register/content	GRAM
0	0	0	0	8080 MCU 8-bit Bus Interface I	D[7:0]	D[7:0],WRX,RDX,CSX,D/CX
0	0	0	1	8080 MCU 16-bit Bus Interface I	D[7:0]	D[15:0],WRX,RDX,CSX,D/CX
0	0	1	0	8080 MCU 9-bit Bus Interface I	D[7:0]	D[8:0],WRX,RDX,CSX,D/CX
0	0	1	1	8080 MCU 18-bit Bus Interface I	D[7:0]	D[17:0],WRX,RDX,CSX,D/CX
0	1	0	1	3-wire 9-bit data Serial Interface I	SCL,SDA,CSX	
0	1	1	0	4-wire 8-bit data Serial Interface I	SCL,SDA,D/CX,CSX	
1	0	0	0	8080 MCU 16-bit Bus Interface II	D[8:1]	D[17:0],D[8:1],WRX,RDX,CSX,D/CX
1	0	0	1	8080 MCU 8-bit Bus Interface II	D[17:10]	D[17:10],WRX,RDX,CSX,D/CX
1	0	1	0	8080 MCU 18-bit Bus Interface II	D[8:1]	D[17:0],WRX,RDX,CSX,D/CX
1	0	1	1	8080 MCU 9-bit Bus Interface II	D[17:10]	D[17:9],WRX,RDX,CSX,D/CX
1	1	0	1	3-wire 9-bit data Serial Interface II	SCL,SDI,SDO,CSX	
1	1	1	0	4-wire 8-bit data Serial Interface II	SCL,SDI,D/CX,SDO,CSX	



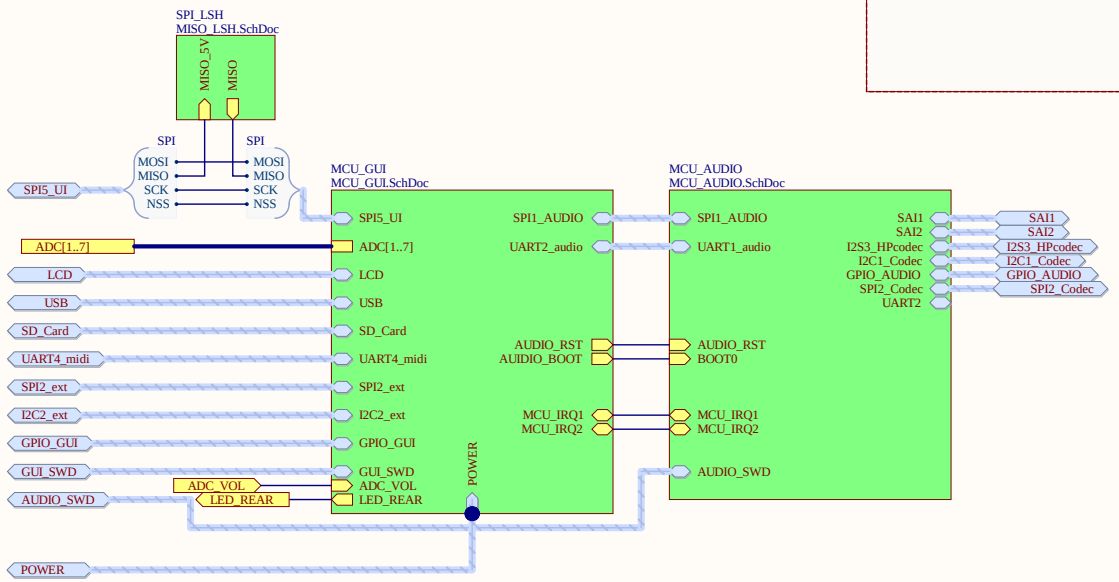
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Size A4	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\..\LCD_big.SchDoc		Drawn By:

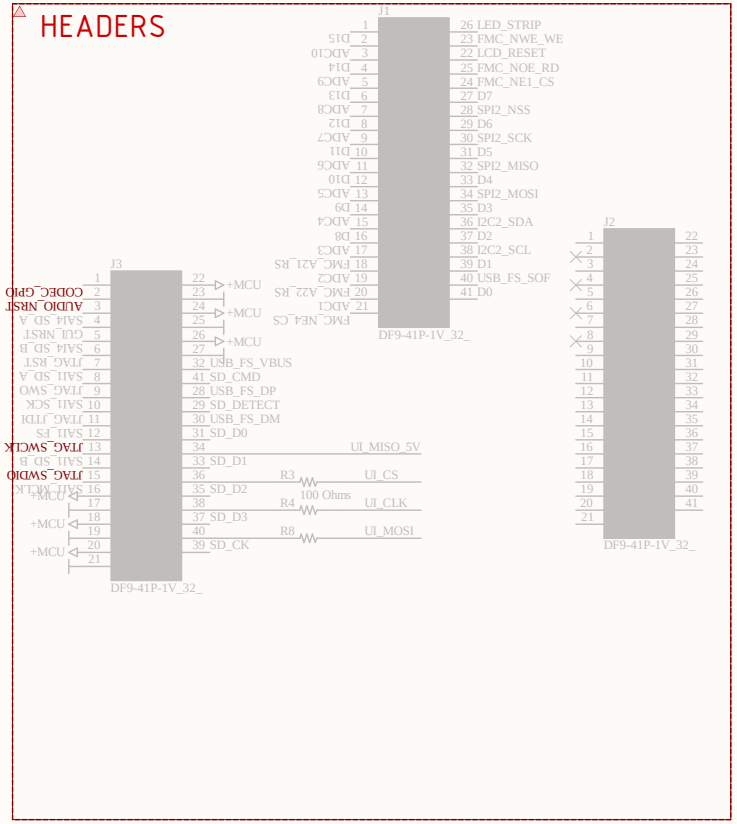


Title		
Size: A4	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\..\FOOT_CN.SchDoc		Drawn By:

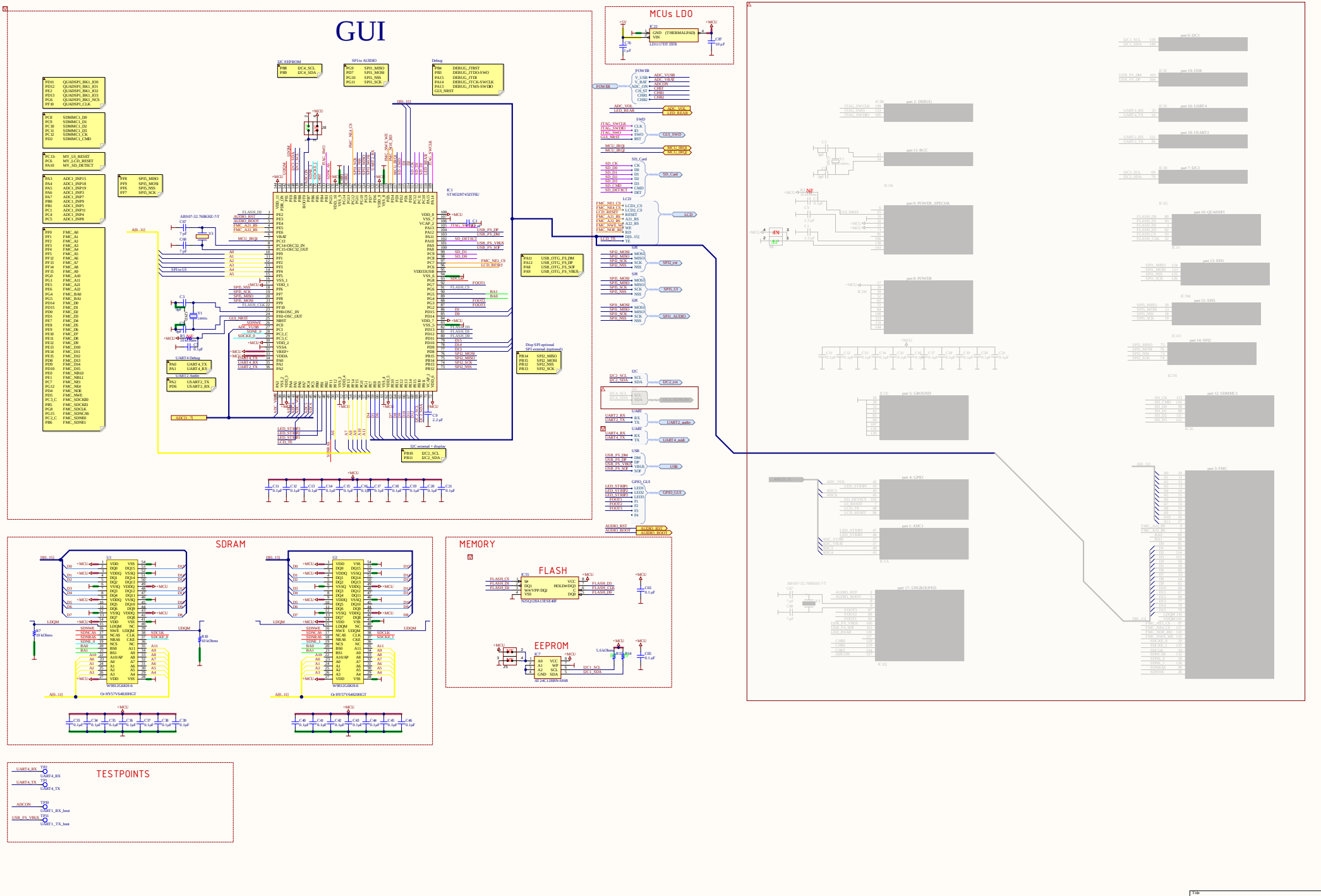


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Size A4	Number	Revision
Date: 10.28.2026		Sheet of
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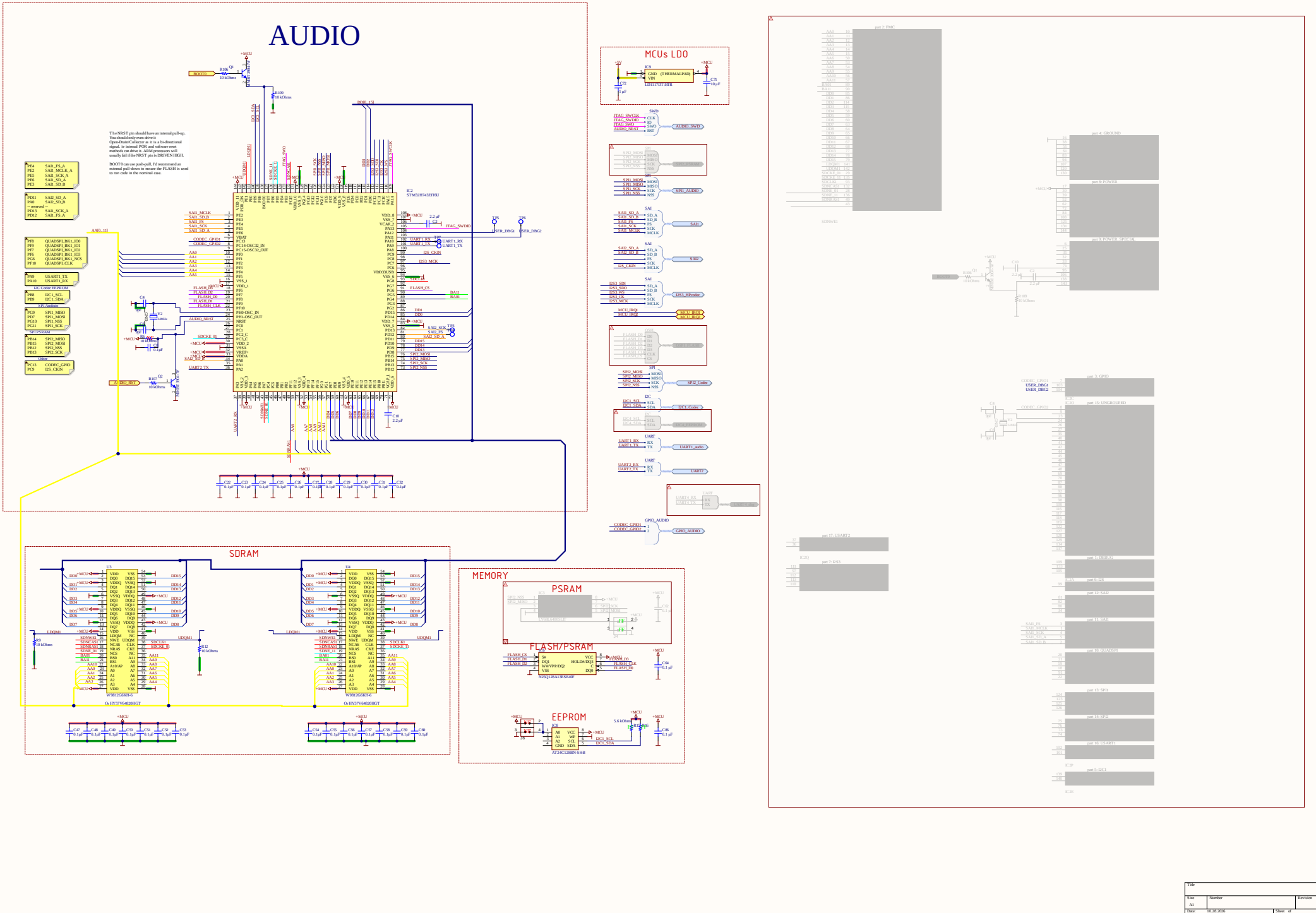




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Size A3	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\..\SOM_MCU.SchDoc		Drawn By:

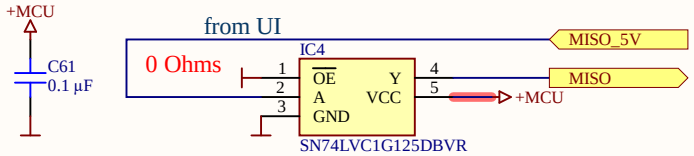

GUI
MCUs LDO
SDRAM
MEMORY
FLASH
EEPROM
TESTPOINTS
POWER
SWD
SD_Card
LCD
QSPI
SPI1_AUDIO
SPI2_ext
SPI5_LS
UART
USB
GPIO_GUI
I2C
I2C2_ext
DISCOVERY
UART4_midi
UART2_audio
AUDIO_RST
AUDIO_BOOT
FLASH_CS
FLASH_D0
FLASH_D1
FLASH_D2
FLASH_D3
FLASH_CLK
EEPROM
I2C1_SCL
I2C1_SDA
UART4_RX
UART4_TX
ADCON
USB_FS_VBUS
UART1_RX_boot
UART1_TX_boot

AUDIO



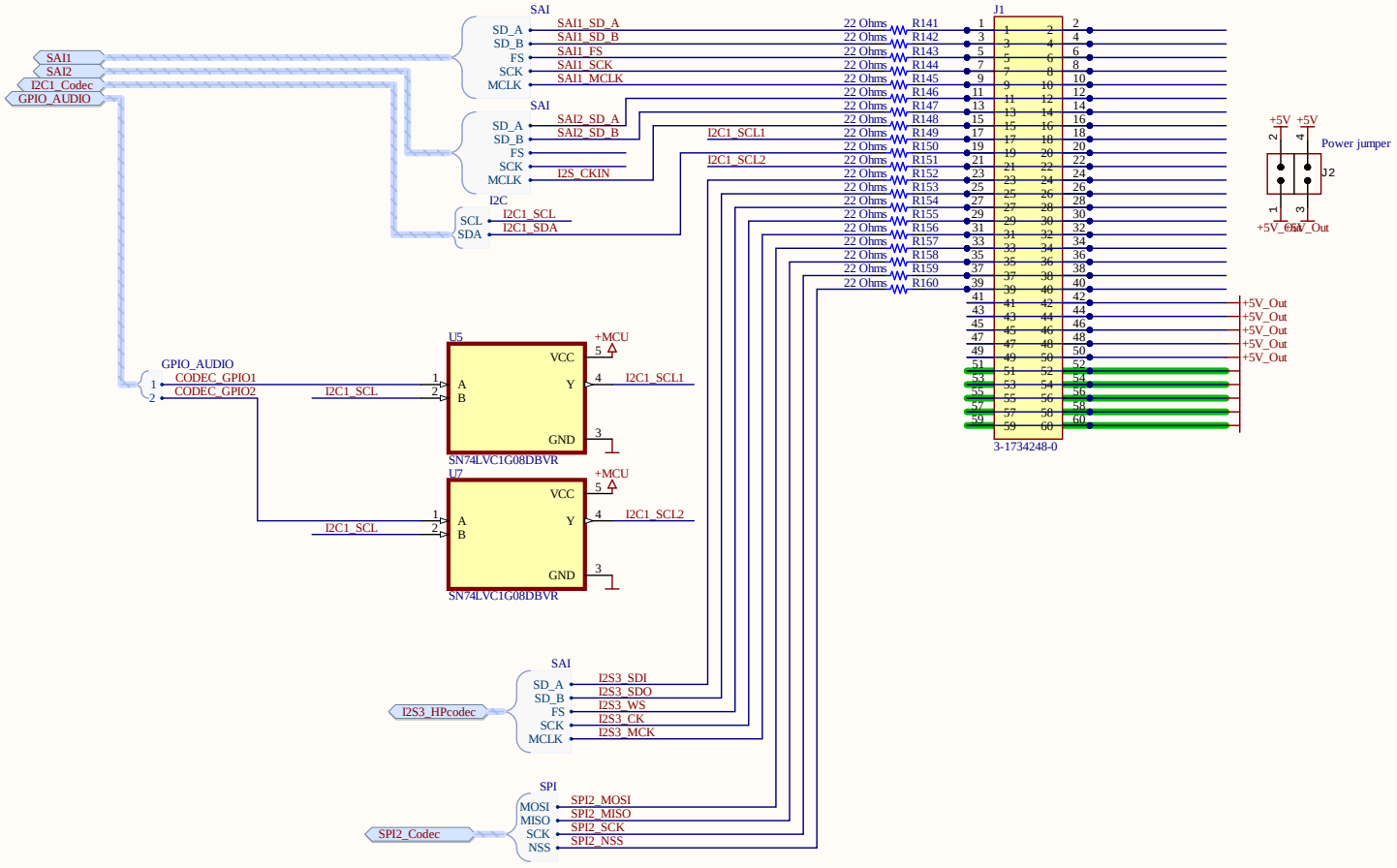
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Size	A1
Number	
Revision	
Date	10.29.2026
Sheet	of
File	C:\Users\...\MCU_AUDIO.SchDoc
Drawn By:	

SPI MISO LEVEL SHIFTER

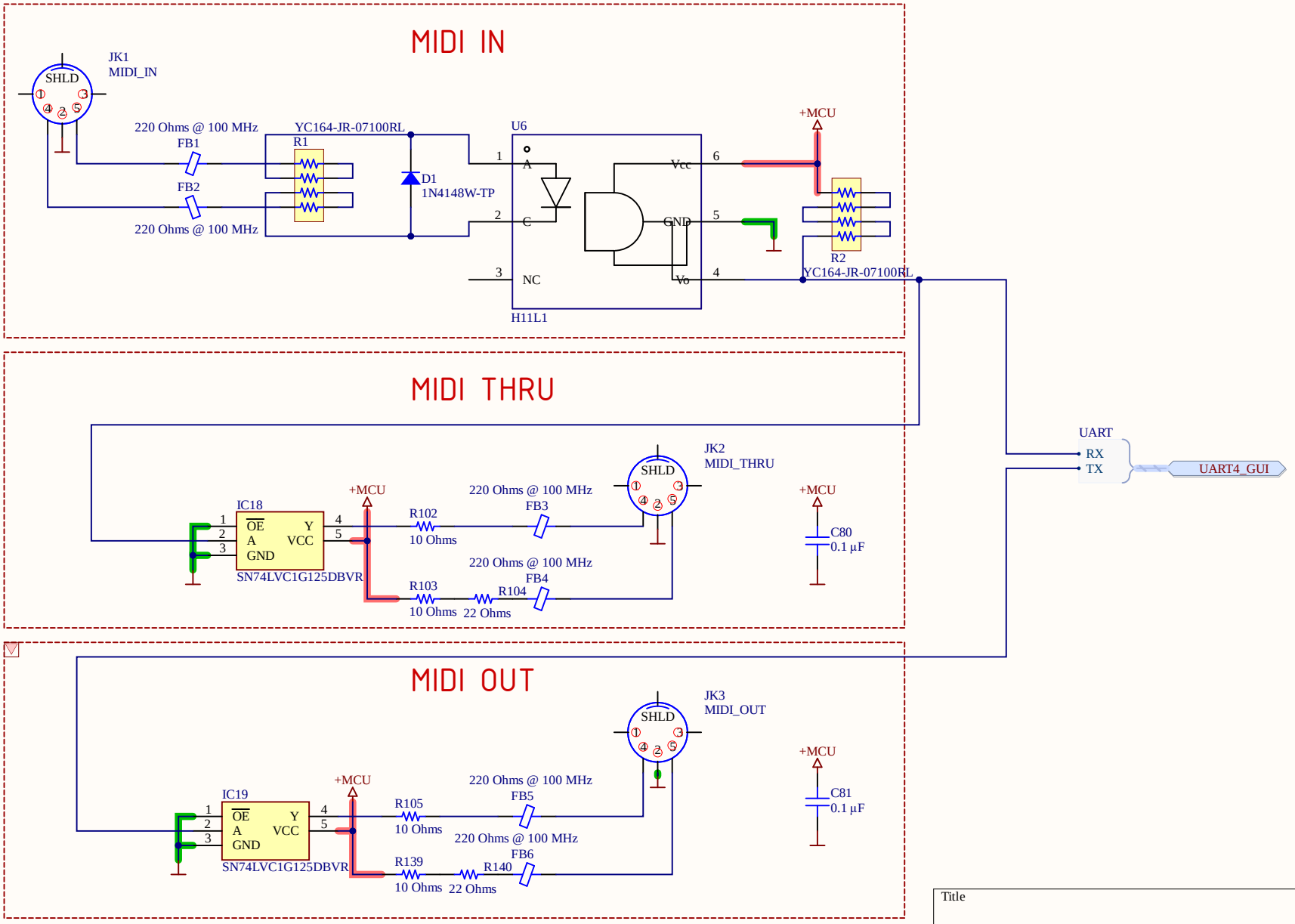


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Date: 10.28.2026		Sheet of
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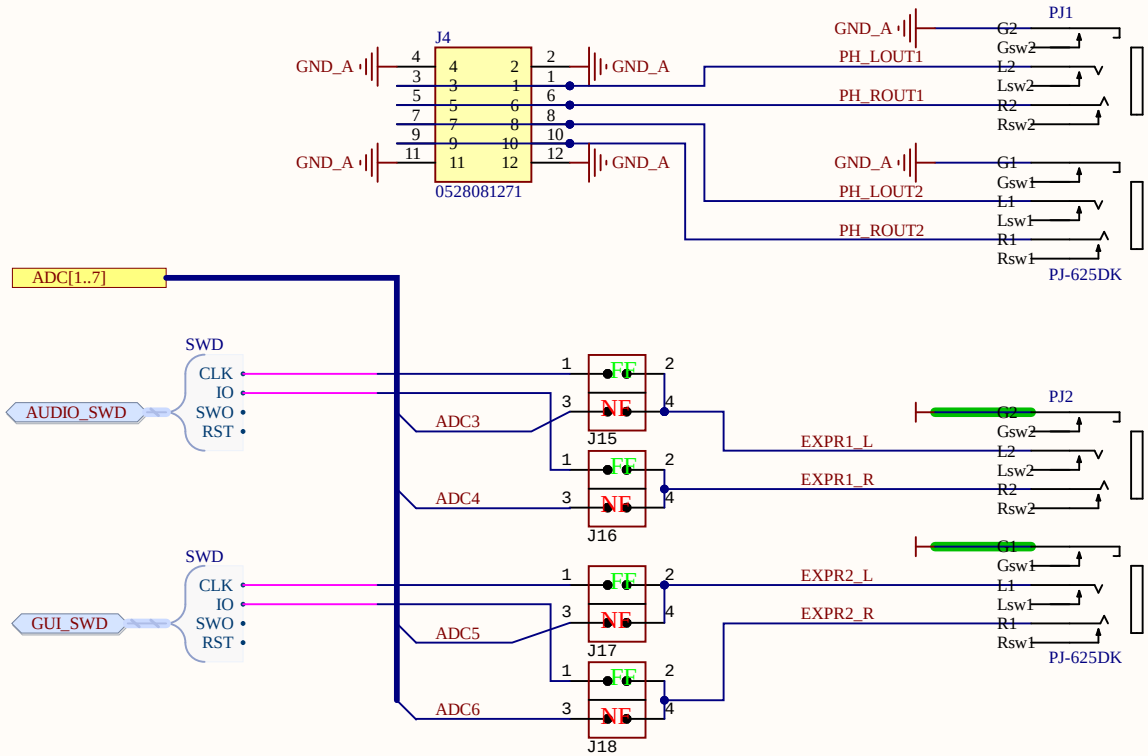
CODEC HEADER



Title		
Size A3	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\..\Codec_CN.SchDoc		Drawn By:

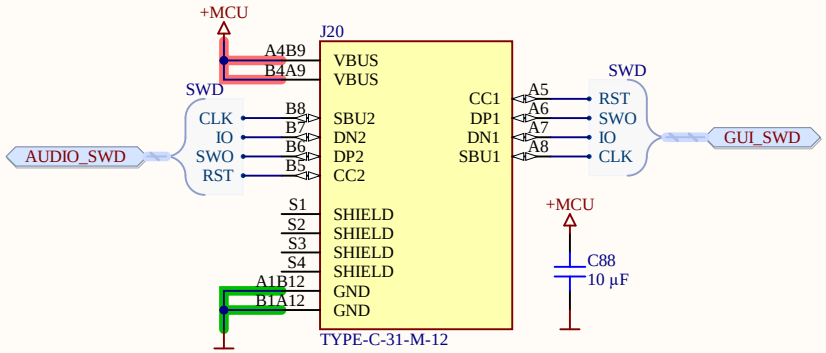


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Size: A4	Number	Revision
Date: 10.28.2026		Sheet of
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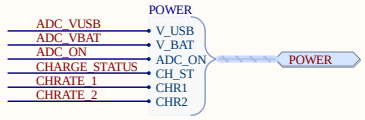


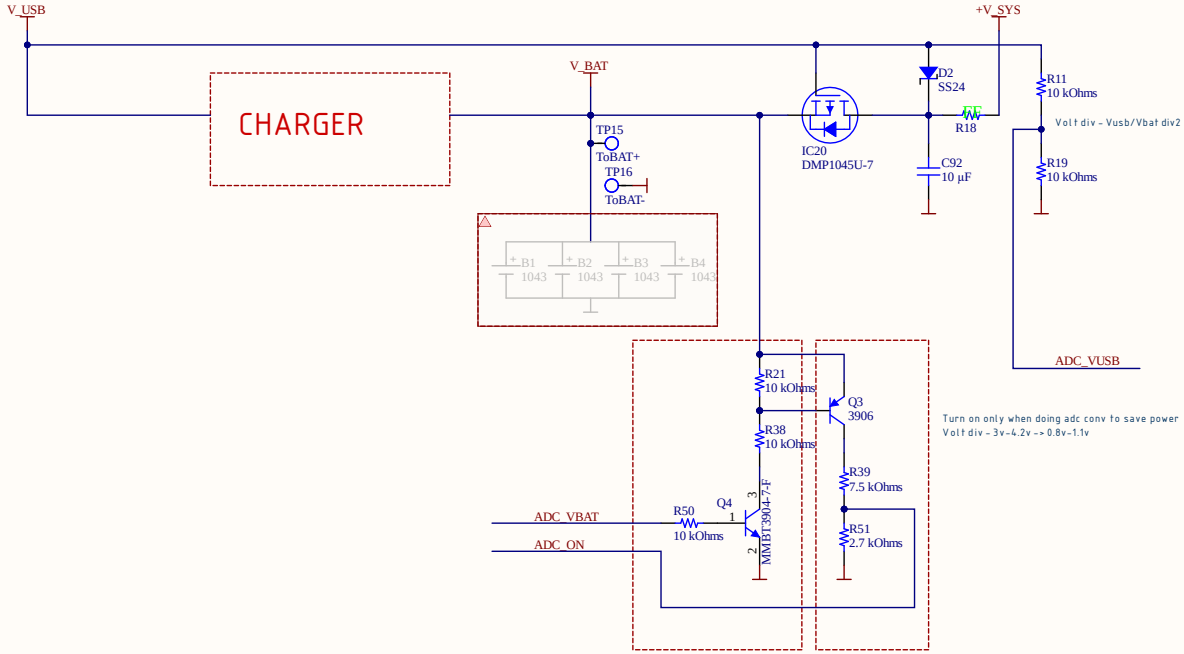
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Size A4	Number	Revision
Date: 10.28.2026		Sheet of
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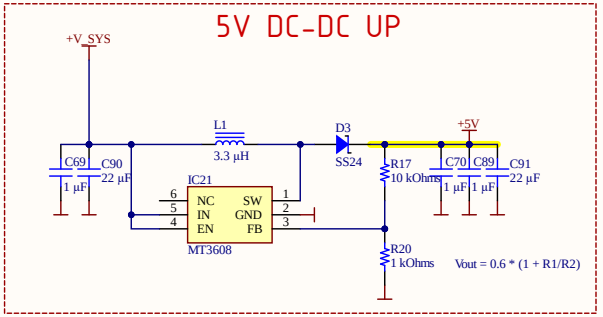
Debug USB

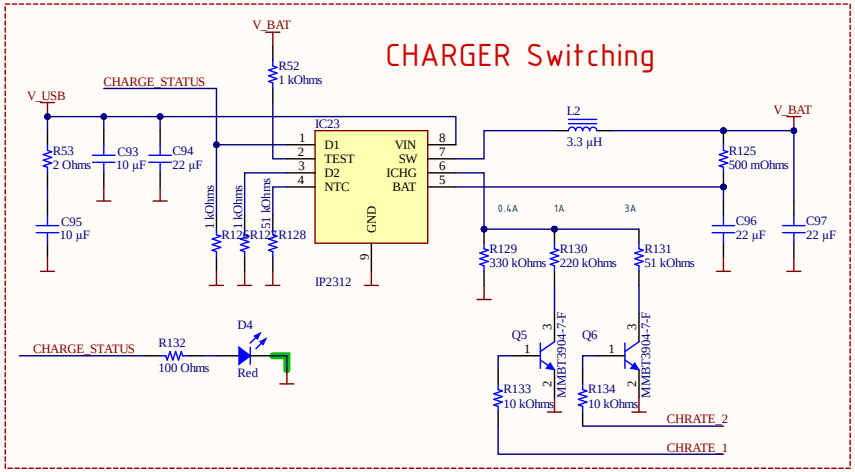


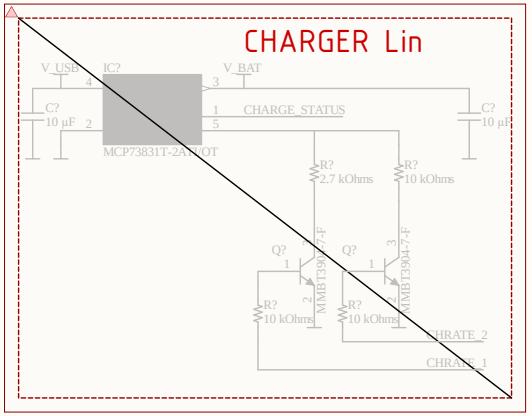
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Size: A4	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\..\USB_DBG.SchDoc		Drawn By:



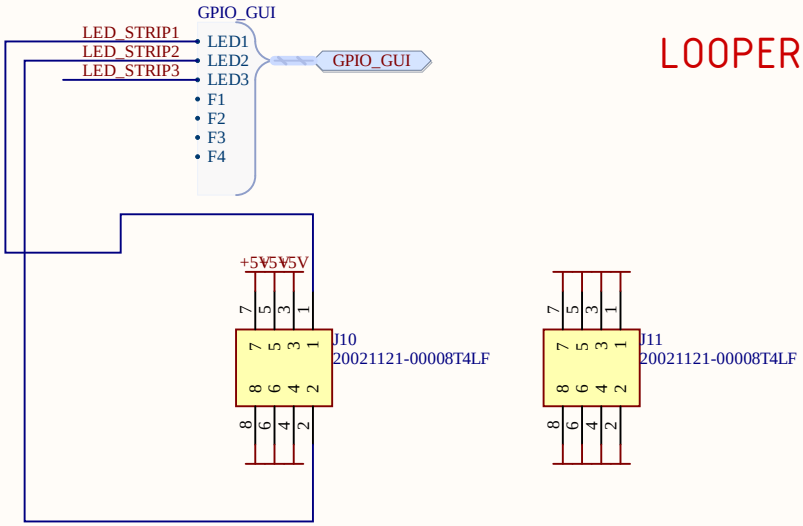






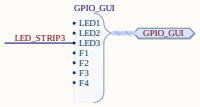


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Size A3	Number	Revision
Date: 10.28.2026		Sheet of
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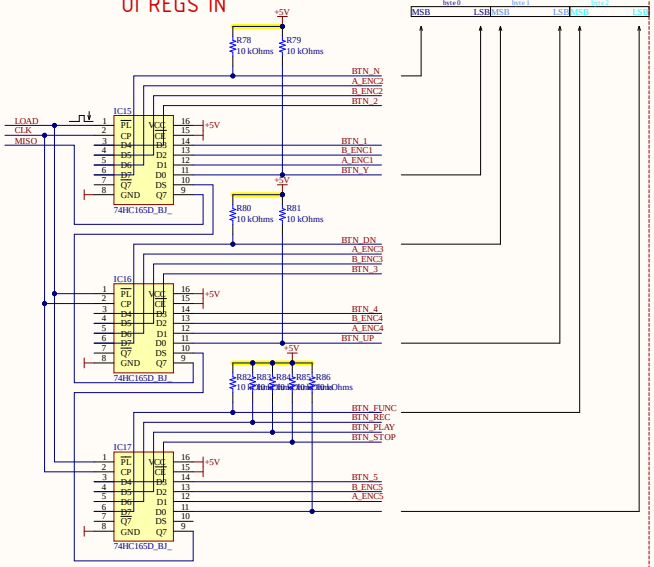


LOOPER LEDs

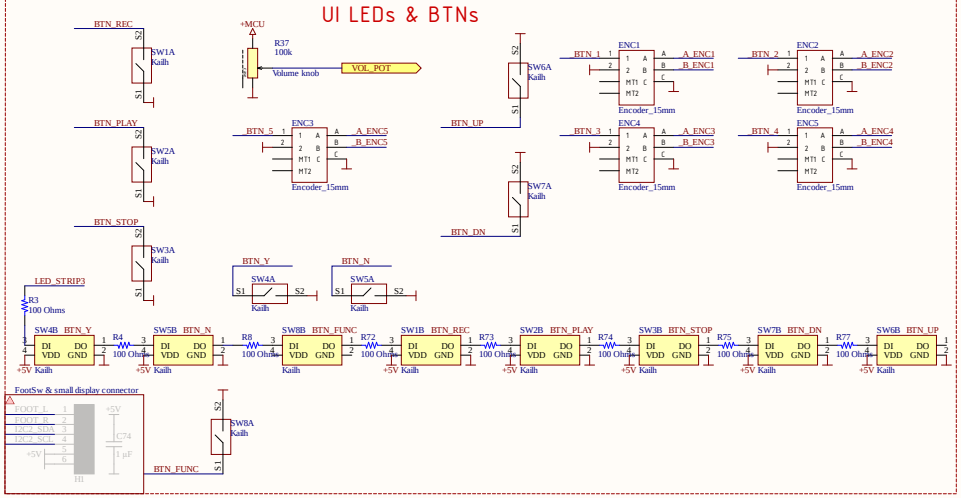
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Size: A4	Number	Revision
Date: 10.28.2026		Sheet of
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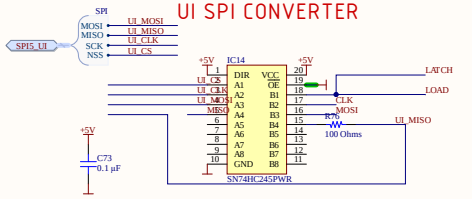
UI REGS IN



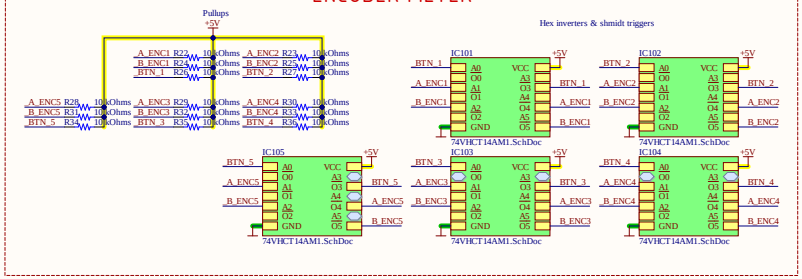
UI LEDs & BTNs



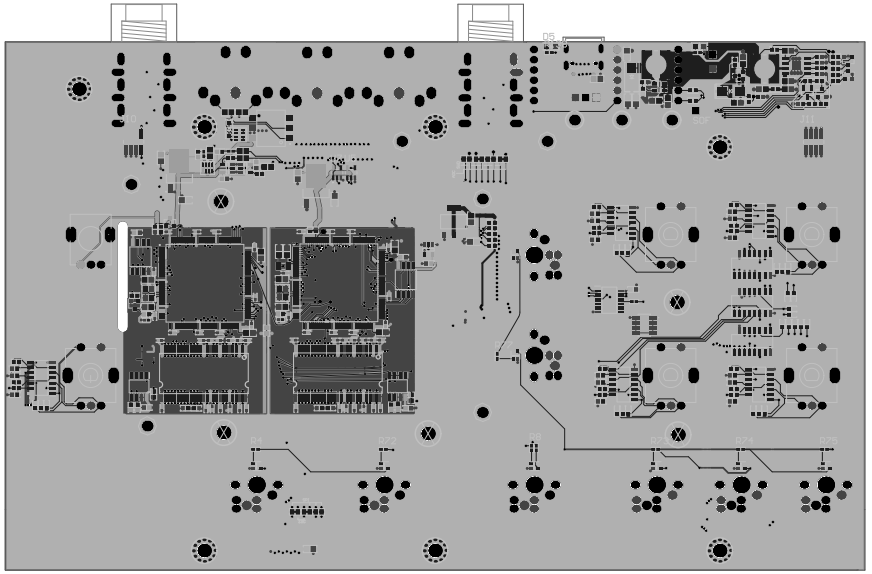
UI SPI CONVERTER



ENCODER FILTER



Title		
Size: A2	Number	Revision
Date: 10.28.2026		Sheet of
File: C:\Users\...\UI.SchDoc		Drawn By:



Line #	Name	Description	Designator	Quantity	Price	Part Number	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1	Supplier 2	Supplier Part Number 2	Supplier Unit Price 2	Supplier Subtotal 2	Manufacturer 2	Manufacturer Part Number 2	Manufacturer Lifecycle 2	Supplier 3	Supplier Part Number 3	Supplier Unit Price 3	Supplier Subtotal 3	Supplier 4	Supplier Part Number 4	Supplier Unit Price 4	Supplier Subtotal 4
	2.2 µF	CAP CER 2.2UF 10V X7R 0805	C1, C2, C9, C10	4	0.03	C0805C225K8RAC7800				DigiKey	399-C0805C225K8RAC7800CT-ND																		
	8pF	CAP CER 8PF 250V C0G/NP0 0805	C3, C4, C5, C6	4	0.28	C0805C080K2GACAUTO				DigiKey	399-8711-1-ND																		
	0.1µF	CAP CER 0.1UF 16V X7R 0402	C7, C8, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60	52	0.01	C0402C104K4RACAUTO				DigiKey	399-6829-1-ND																		
	0.1 µF	CAP CER 0.1UF 50V X7R 0603	C603, C63, C64, C65, C73, C77, IC101, C77, IC102, C77, IC103, C77, IC104, C77, IC105, C78, IC201, C78, IC102, C78, IC103, C78, IC104, C78, IC105, C79, IC201, C79, IC202, C79, IC203, C79, IC204, C79, IC205, C80, C81, C88	23	0.01	C0603C104K5RAC7411				DigiKey	399-C0603C104K5RAC7411CT-ND																		
	7pF	CAP CER 7PF 50V C0G/NP0 0805	C87, C88	2	0.01	CC0805DRNP09BB070				DigiKey	311-1095-1-ND																		
	1 µF	CAP CER 1UF 16V X7R 0805	C69, C70, C72, C75, C76, C89	6	0.01	C0805C105K4RAC7800				DigiKey	399-C0805C105K4RAC7800CT-ND																		
	10 µF	CAP CER 10UF 10V X5R 0805	C71, C74, C85, C87, C86, C90, C83, C95	8	0.02	C0805C106K8PAC7800				DigiKey	399-C0805C106K8PAC7800CT-ND																		
	33 uF	CAP CER 22UF 25V X5R 0805	C90, C91, C94, C96	4	0.09	CC0805MKX5R8BB336				DigiKey	311-1363-1-ND																		
	1N4148W-TP	DIODE GEN PURP 100V 150MA SOD123	D1	1	0.01	1N4148W-TP				DigiKey	1N4148WTPMSCT-ND																		
	B224	DIODE SCHOTTKY 40V 2A DO214AA	D2, D3	2	0.10	B224				DigiKey	B240FSCT-ND																		
	Red	LED RED CLEAR 0805 SMD	D4, D5	2	0.11	LSM380J5431V-CT-ND				DigiKey	SN-LSM380J5431V-CT-ND																		
	Encoder_15mm		ENC1, ENC2, ENC3, ENC4, ENC5	5																									
	220 Ohms @ 100 MHz	FERRITE BEAD 220 OHM 0805 1LN	FB1, FB2, FB3, FB4, FB5, FB6	6	0.02	BLM21PG221SN1D				DigiKey	490-1054-1-ND																		
	testpins_header_6		H1, H2, H4	3																									
	SRC_50_OBC		H3	1																									
	LSS_XGB		H7	1																									
	SN65HVD1427DR	Integrated Circuit	IC1, IC2	2																									
	SN74LVC1G125DBVR	Integrated Circuit	IC4, IC18, IC19	3																									
	AO3413/BA3/SE4/B	Integrated Circuit	IC5, IC6	2																									
	M24C128BW-SH-B	Integrated Circuit	IC7, IC8	2																									
	L3111-TOT1078	Integrated Circuit	IC9, IC23	2																									
	ams1117-3.3V	Integrated Circuit	IC10	1						LCSC	C6186																		
	74VHCT14AM	Integrated Circuit	IC11, IC101, IC11_IC102, IC11_IC103, IC11_IC104, IC11_IC105	5																									
	SN74HC245PWR	Integrated Circuit	IC14	1																									
	74VHC1G50_IEL	Integrated Circuit	IC15, IC16, IC17	3																									
	SMP1045U-7	MOSFET P-CH 12V 4A SOT23	IC20	1	0.07	SMP1045U-7DICT-ND				DigiKey	SMP1045U-7-DICT-ND																		
	MP3608	Integrated Circuit	IC21	1																									
	PT2313	Integrated Circuit	IC22	1																									
	5-1734248-0	Connector	J3	1																									
			J2, J5, J6, J7, J8, J9, J14, J15, J16, J17, J18, J19	12																									
	5023690127	Connector	J3_J4	1																									
	53261123-0000974-F	Connector	J20, J21	2																									
	TYPE-C-31-M-12	USB Connectors 24 Receptacle 1 8.94×7.3mm RoHS	J30	1	None																								
	MIDI_IN	SINS	JA1	1																									
	MIDI_THRU	SINS	JA2	1																									
	MIDI_OUT	SINS	JA3	1																									
	6.8 µH	FIXED IND 3.3UH 2.5A 28 MOHM SMD	L1, L2	2	0.31	SRR6028-3R3Y				DigiKey	1116-SRR6028-3R3YCT-ND																		
	MountingHole		MH1, MH2, MH3, MH4, MH5, MH6, MH7, MH8	8																									
	IO_EDGE	0.3mm jack HCO/YA	P1, P2	2																									
	MMBT3904-7-F	TRANS NPN 40V 0.2A SOT23-3	Q1, Q2, Q4, Q5, Q6	5	0.01	MMBT3904-7-F				DigiKey	MMBT3904-FDICT-ND																		
	SW6		Q3	1																									
	YC164-JR-071K8L	RES ARRAY 4 RES 10Ω OHM 1206	R1, R2	2	0.01	YC164-JR-071K8L				DigiKey	YC164J-10CT-ND																		
	100 Ohms	RES SMD 100 OHM 1% 1/16W 0402	R3, R4, R8, R72, R73, R74, R75, R76, R77, R110, R138	11	0.01	CRCW040210R0FKEE				DigiKey	541-29E3-1-ND																		
	10 kOhms	RES SMD 10K OHM 1% 1/10W 0603	R5, R6, R7, R9, R10, R11, R12, R17, R19, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R38, R46, R47, IC201, R47, IC102, R47, IC103, R47, IC104, R47, IC105, R48, IC101, R48, IC102, R48, IC103, R48, IC104, R48, IC105, R49, IC101, R49, IC102, R49, IC103, R49, IC104, R49, IC105, R50, R78, R79, R80, R81, R82, R83, R84, R85, R86, R106, R107, R108	57	0.01, [NoParam]	ERJ-3EKF1002V				DigiKey [NoParam]	P10.0KHCT-ND, [NoParam]															
	5.6 kOhms	RES SMD 5.6K OHM 0.5% 1/10W 0603	R13, R14, R15, R16	4	0.01	RT0603DRD075561				DigiKey	311-2635-1-ND																		
	Value	RES SMD 0 OHM JUMPER 1/10W 0603	R18	1	0.01	RCA06030000Z0EA				DigiKey	541-2192-1-ND																		
	1 kOhms	RES SMD 1K OHM 1% 1/10W 0603	R20, R52, R127	3	0.01	CRCW06031k00FKEB				DigiKey	541-2985-1-ND																		
	SW6		R37	1																									
	7.5 kOhms	RES 7.5K OHM 1% 1/10W 0603 SMD	R39	1	0.01	ERJ-U03F7501V				DigiKey	10-ERJ-U03F7501VCT-ND																		
	22 Ohms	RES 22 OHM 5% 1/16W 0402	R40, R41, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R104, R137, R140, R141, R142, R143, R144, R145, R146, R147, R148, R149, R150, R151, R152, R153, R154, R155, R156, R157, R158, R159, R160	43	0.01	RMCF0402JT22R0CT-ND				DigiKey	RMCF0402JT22R0CT-ND															
	10 Ohms	RES SMD 10 OHM 1% 1/3W 0603	R42, R43, R44, R45, R102, R103, R105, R139	8	0.01	ERJ-PA3F10R0V				DigiKey	P10.00BHCT-ND																		
	2.7 kOhms	RES 2.7K OHM 1% 1/10W 0603	R51	1	0.01	CRCW06032K70FKEA				DigiKey	541-5328-1-ND																		
	2 Ohms	RES 2 OHM 1% 1/10W 0603 SMD	R53	1	0.01	ERJ-U03J2R0V				DigiKey	10-ERJ-U03J2R0VCT-ND																		
	500 mOhms	RES 0.5 OHM 1% 1/10W 0603	R125	1	0.11	RLT0603-C-R500-FNH				DigiKey	2037-RLT0816-C-R500-FNHCT-ND																		
		RES SMD 1K OHM 1% 1/10W 0603	R126	1	0.01	CRCW06031k00FKEB				DigiKey	541-2985-1-ND																		
	31.6 kOhms	RES 31K OHM 1% 1/10W 0603	R128, R131	2	0.01	RMCF0603FT31K6				DigiKey	RMCF0603FT31K6CT-ND																		
	330 kOhms	RES 330K OHM 5% 1/10W 0603	R129	1	0.01	RMCF0603JG330KCT				DigiKey	RMCF0603JG330KCT-ND																		
	220 kOhms	RES 220K OHM 1% 1/10W 0603	R130	1	0.01	CRCW06032203FKEA				DigiKey	541-4039-1-ND																		
	Value	RES SMD 0 OHM JUMPER 1/10W 0603	R135, R136	2	0.01	RCA06030000Z0EA				DigiKey	541-2192-1-ND																		
	Keith	Keith keyboard switch, hotswap rgb, dual-side hotswap options + low profile variant	SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8	8																									
	SJ4874_TX		SJ1	1																									
	SJ4874_RX		TP2	1																									
	SAO_2CL		TP3	1																									
	SAO_2L		TP4	1																									
	USB_D6G1		TP5	1																									
	USB_D6G2		TP6	1																									
	SJ4873_RX		TP7	1																									
	SJ4873_TX		TP8	1																									
	5V2		TP9	1																									
	SJ4871_BX_boot		TP10	1																									
	SJ4871_TX_boot		TP11	1																									
	ToPower		TP12, TP13, TP14	3																									
	ToBAT+		TP15	1																									
	ToBAT-		TP16	1																									
	W981ZG6KI-6		U1, U2, U3, U4	4																									
	SN74LVC1G08DBVR	1-ch, 2-input 1.65-V to 5.5-V 32 mA drive strength AND gate	U5, U7	2																									
	SSJ3-1	OPTOISO 4.17KV OPN OCSL 8SMD	U6	1																									
	24MHz	CRYSTAL 24MHZ 9PF SMD	Y1, Y2	2	0.39	9X32250A-24MHC9				DigiKey	644-1363-1-ND																		
	ABM3T-32.768KHZ-7-T	Crystal or Oscillator	Y3	1		EX205A-C3D7530																							